Soumyajyoti Dutta

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Applied AI Deployment | RAG & Agentic Systems | LLM Fine-Tuning & Evaluation

Expertise

Core Competencies:

- RAG system design: retrieval pipeline architecture, confidence-threshold anti-hallucination, live web-search fallback integration, and end-to-end deployment on local GPU infrastructure.
- LLM fine-tuning and deployment: curriculum learning, structural reinforcement, and reward signal design across 7+ model families (140M–26B parameters); 150+ models trained, evaluated, and deployed.
- Agentic system development: multi-tool dispatcher architecture (20 tools), 4-layer query routing, synthesizer integration, subprocess isolation, and production hardening.
- Evaluation frameworks: custom reward metrics (lexical + syntactic + semantic), LLM-as-judge simulation, and human-curated benchmarking — applied at both training time and production quality gates.
- Technical communication and client delivery: explaining ML systems to mixed-background audiences; client-embedded consulting experience driving stakeholder alignment across engineering and product teams.

Technical Stack:
Python (PyTorch, HuggingFace Transformers, TRL, TensorFlow, scikit-learn, LightGBM, NumPy, Pandas), DeepSpeed, FSDP, DDP/NCCL, CUDA, SLURM, C++, Rust, AWS, Git, CI/CD, Django/Flask, YARA

Professional Experience

Texas A&M University — College Station, TX
Research Assistant (Affiliation: Botacin's Lab) — **Spring 2024 – Present**

- Built and deployed an end-to-end **threat intelligence agent** on a local GPU cluster: 20-tool dispatcher with a 4-layer query router, **RAG pipeline** (curated malware knowledge base, confidence-threshold anti-hallucination, live Serper web-search fallback, clarification prompting on low-confidence queries), and Nemotron-4B synthesizer.
- Fine-tuned **7+ LLM families** (BART, T5, Nemotron 3 Nano 9B, Gemma 3 270M/1B/4B, LLaMA 3.1 8B, LLaMA 3.2 1B/3B) on 100M+ labeled examples using curriculum learning and structural reinforcement; **150+ models trained and evaluated** on multi-node SLURM clusters.
- **YaraGen** (USENIX Security 2026 submission in preparation):
  - ∗ Designed YaraAST hybrid tokenizer — captures YARA grammar at the atom level, reducing token space by 10×; trained BART-base 140M from scratch achieving **99% syntax validity** and **95% rule correctness** on 22,655 held-out rules.
  - ∗ Engineered Scooper (BiLSTM hex-sequence extractor with typed placeholder reinsertion) and MultiRuleGen (deterministic IR-level merger) to handle 17% of rules exceeding the 1,024-token window — raising corpus coverage from 83% to **100%**.
  - ∗ Designed *Brownie & Puff*: evaluation function (BLEU + Parsing + SAT semantics) used as training-time reward signal and production quality gate; benchmarked across T5, BART, and LLaMA families.
  - ∗ Built DeBERTa-v3 + dual CNN/Transformer char-encoder NER pipeline: **92.5% F1** across 25 IOC types on real analyst reports (CISA, NCSC, ESET, Cisco Talos, Unit 42).
- **AutoPYara** (open-sourced on PyPI, ACSAC 2026 submission):
  - ∗ Biclustering-based YARA rule generation from raw malware binaries (71K Windows PE samples); +14% rule coverage, +10% threat hunting accuracy vs. AutoYara SOTA.

Cognizant Technology Solutions — Kolkata, India
Junior Software Engineer (Consulting) — **2021 – 2022**

- Client-facing technical consultant embedded with **TJX Companies**; coordinated daily with client engineering and product stakeholders to scope, build, and deliver a logistics automation platform.
- Optimized REST API endpoints for latency and concurrency; maintained CI/CD pipelines and led code review processes across cross-functional delivery teams.

Projects

Ambiguity-Aware Deep Research Agent | *Independent Research* 2026

- Fine-tuned Gemma-4-26B-A4B (sparse MoE: 4B active / 26B total) on 16 H100 GPUs via DeepSpeed and FSDP to build a research agent that detects ambiguous queries and asks targeted follow-up questions rather than producing broad, hallucination-prone responses.
- Three-stage curriculum: (1) ambiguity identification, (2) confidence rating with chain-of-thought reasoning, (3) targeted follow-up question generation.
- Evaluated via GPT-4o user-simulation framework against a 4-person human-curated ground-truth test set; results: reduced token usage, lower hallucination rate, and improved user satisfaction vs. baseline agent.

Machine Learning Malware Detection Model | *1st Place*

- End-to-end ML system for Windows PE classification; integrated EMBER, BODMAS, and Benign-NET datasets (2M+ samples); LightGBM/XGBoost achieving 96.2% accuracy, 0.97 F1.

HelloPentagon

- Explainable ML malware defense chatbot integrating a classifier with LLM-based triage and interpretability — deployed as an interactive tool for security analysts.

Teaching & Outreach

Teaching Assistant (CSCE 439/704) Texas A&M University
Data Analytics for Cybersecurity **Spring 2026**

- Explained clustering, supervised ML, and anomaly detection to a mixed-background cohort of graduate and advanced undergraduate students; developed assignments and provided hands-on project mentorship.

Partner & Presenter Texas A&M University
TAMU CS Day **2024, 2025**

- Conducted interactive AI and cybersecurity demos for high school students — translating research-level concepts for a general audience.

Education

Texas A&M University College Station, TX
Ph.D. in Computer Science — Advisor: Dr. Marcus Botacin **Spring 2024 – Present**

Texas A&M University College Station, TX
M.S. in Computer Engineering **Fall 2022 – Fall 2023**

SRM University Chennai, India
B.Tech. in Electronics and Communication Engineering **Fall 2017 – Spring 2021**